Case Study 10: Weekly Electricity Consumption Matrix

1. Problem Statement

Record electricity consumption for multiple appliances over seven days. Calculate daily totals and appliance-wise totals, then identify consumption above a specified limit.

2. Objectives

Use nested lists for appliance-wise daily readings.

Calculate row-wise and column-wise totals.

Identify readings or totals that exceed a limit.

3. Assumptions

There are three appliances and seven days.

All consumption readings are non-negative kWh values.

The daily limit is entered by the user and applies to the total consumption for each day.

4. Algorithm

1. Start.

2. Input names of three appliances and seven daily readings for each.

3. Calculate each appliance’s weekly total.

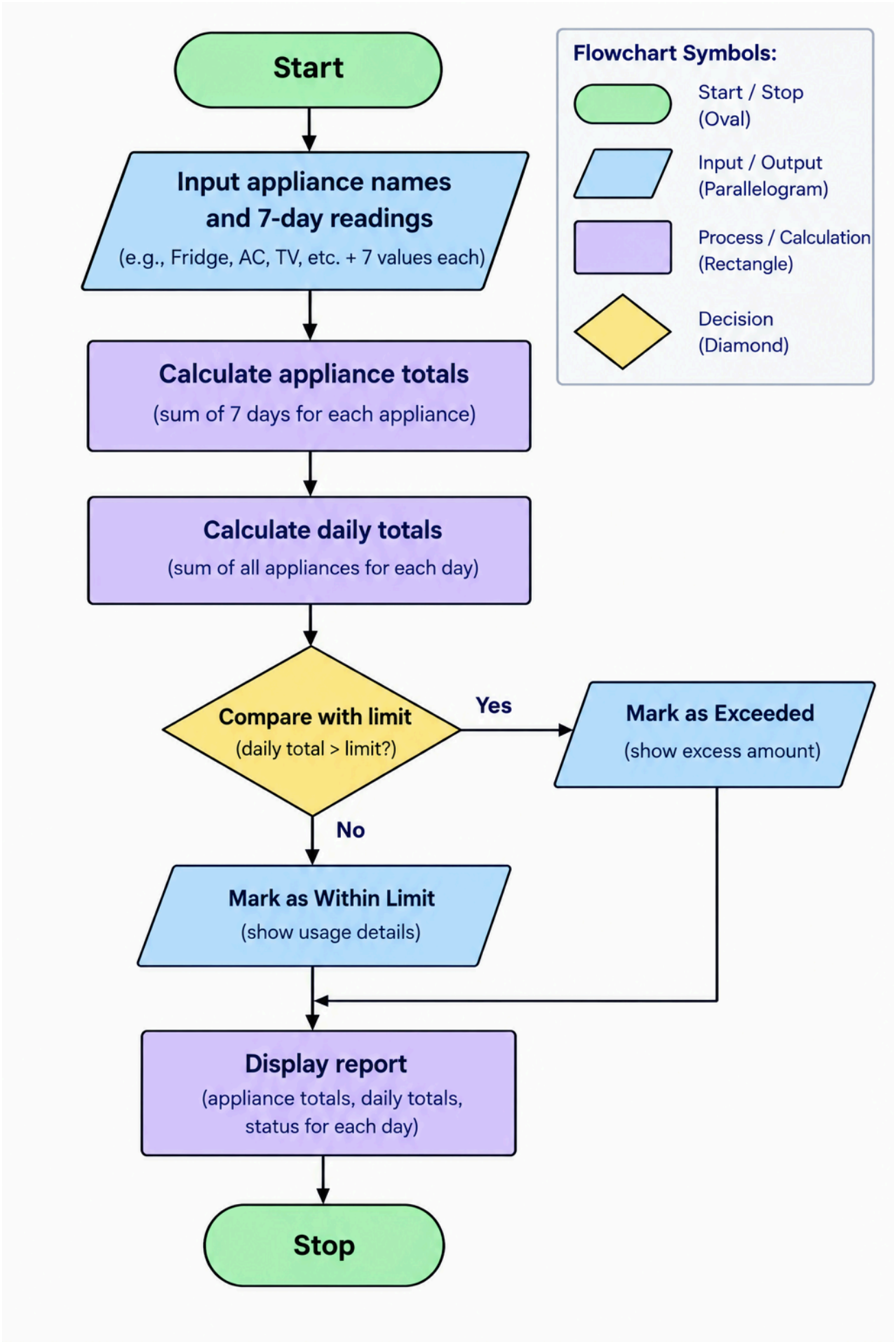
4. Calculate the total for each day.

5. Compare daily totals with the entered limit.

6. Display weekly totals and days exceeding the limit.

7. Stop.

5. Flowchart (text form)


Start
Input appliance names and 7-day readings
(e.g., Fridge, AC, TV, etc. + 7 values each)
Calculate appliance totals
(sum of 7 days for each appliance)
Calculate daily totals
(sum of all appliances for each day)
Compare with limit
(daily total > limit?)
Yes
Mark as Exceeded
(show excess amount)
No
Mark as Within Limit
(show usage details)
Display report
(appliance totals, daily totals, status for each day)
Stop
Flowchart Symbols:
Start / Stop (Oval)
Input / Output (Parallelogram)
Process / Calculation (Rectangle)
Decision (Diamond)

6. Pseudocode

START
INPUT daily_limit
FOR each of 3 appliances: INPUT name and 7 readings
FOR each appliance: calculate sum of its 7 readings
FOR each day: add readings from all appliances
IF daily total > daily_limit: flag that day
DISPLAY appliance totals, daily totals, and flagged days
STOP

7. Complete Python Source Code

```
appliances = []
usage_matrix = []

daily_limit = float(input("Enter daily consumption limit in kWh: "))
```



```
for i in range(3):
 name = input(f"Enter appliance {i + 1} name: ")
readings = []
 for day in range(7):
usage = float(input(f"Enter {name} usage for day {day + 1} (kWh): "))
 while usage < 0:
 print("Usage cannot be negative.")
usage = float(input(f"Re-enter {name} usage for day {day + 1}: "))
readings.append(usage)
appliances.append(name)
usage_matrix.append(readings)

print("
----- APPLIANCE WEEKLY TOTALS -----")
for i in range(3):
 print(f"{appliances[i]}: {sum(usage_matrix[i]):.2f} kWh")

print("
----- DAILY TOTALS -----")
for day in range(7):
daily_total = 0
 for appliance in range(3):
daily_total += usage_matrix[appliance][day]

 print(f"Day {day + 1}: {daily_total:.2f} kWh", end="")
 if daily_total > daily_limit:
 print(" - EXCEEDS LIMIT")
 else:
 print(" - Within limit")
```

8. Test Case and Expected Output

Example readings: Fan uses 1 kWh each day, light uses 0.5 kWh each day, and computer uses 2 kWh each day. Daily limit = 3 kWh.

Fan: 7.00 kWh Light: 3.50 kWh Computer: 14.00 kWh Each day total: 3.50 kWh - EXCEEDS

LIMIT **9. Conclusion**

The program analyses a weekly consumption matrix, calculates appliance and daily totals, and flags days exceeding the

specified limit.